



MEDREACH

Medreach

The right hospital, right now –
confirmed before you leave home

AI-powered emergency coordination for **Madhya Pradesh**, connected to the **108 / 102 national ambulance services**.



Speak in Hindi or English
AI understands the emergency



Hospital accepts first
no wasted trips, ER is ready



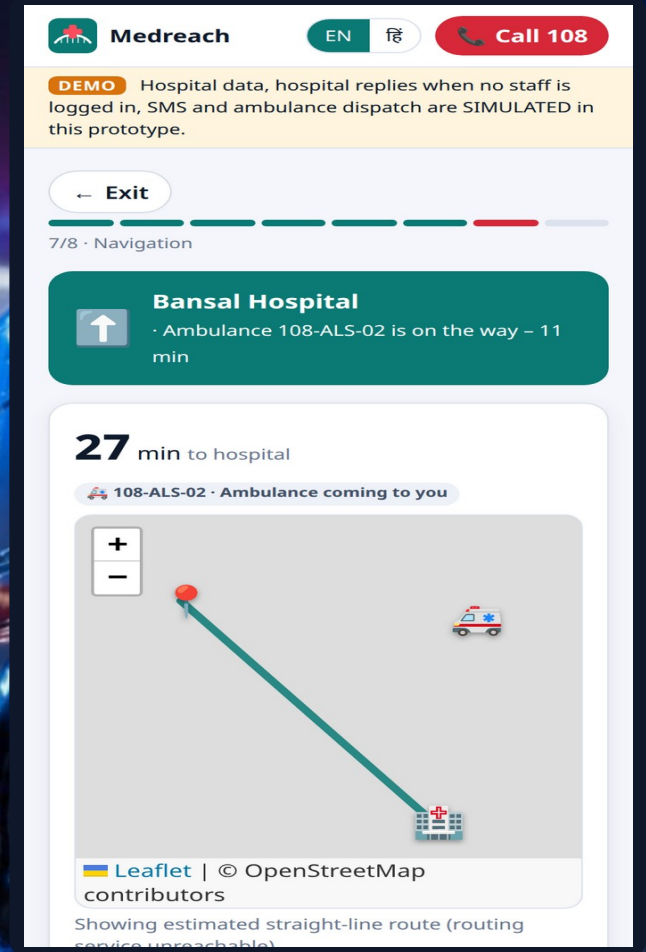
108 ambulance dispatched
nearest unit, tracked live



Works on 2G, SMS, offline
built for villages

MPOnline Idea & Innovation Hackathon 2026 • Challenge 5: Healthcare • Team Medreach

Live: medreach-2.onrender.com | github.com/prakharporwal518-collab/Medreach



9:40 PM, a Village Near Berasia: Chest Pain

Ramesh's father (58) collapses with chest pain and sweating. Only 2G works. The nearest town is 25 km away. What happens next decides his life.

⊗ Today

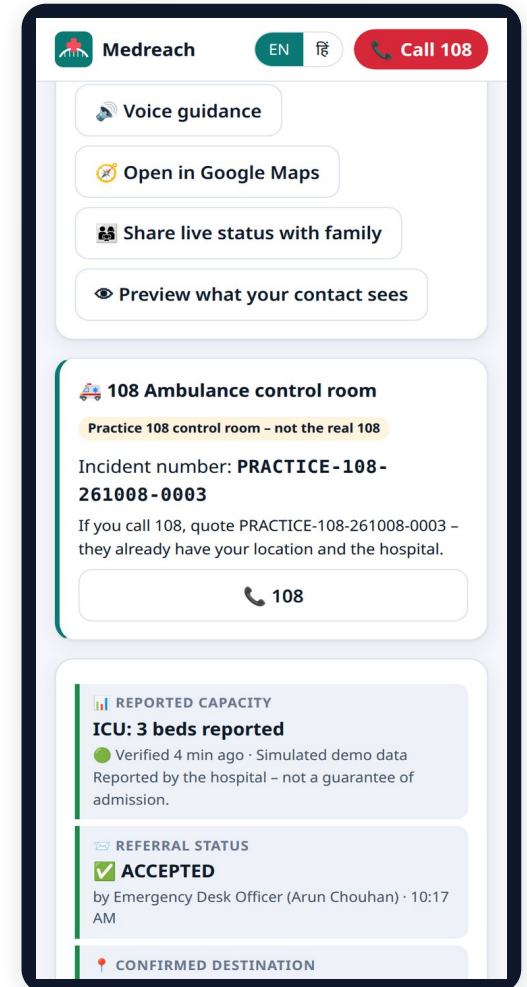
- 0 min** Panic. Calls relatives – nobody knows which hospital has a cardiologist tonight.
- +20 min** Drives to the nearest hospital: no cardiologist, ICU full.
- +45 min** Calls 108 and repeats the address and details; the weak 2G call drops.
- +90 min** Reaches a second hospital that was not expecting him.

The golden hour is lost on the road.

✓ With Medreach

- 0:00** Son speaks in Hindi: “papa ko seene mein dard hai...”
- 0:30** AI: possible heart attack – CRITICAL. Needs cardiologist + ICU.
- 1:00** Only hospitals that can treat it, ranked by match score.
- 2:00** Bansal Hospital accepts and reserves a resus bay.
- 2:30** 108 incident created, nearest ALS ambulance sent and tracked.

He arrives at an ER that is ready for him.



What the family sees: hospital accepted + 108 incident number.

One Platform for Citizen, Hospital and 108

THE PROBLEM

1,68,491

road-accident deaths in India in 2022

Source: MoRTH, Road Accidents in India 2022



Nearest ≠ capable

maps show distance, not the specialist on duty



No trusted live status

bed figures are old, nobody owns them



No acceptance first

refusal happens at the hospital gate



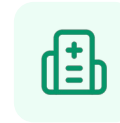
Weak networks

villages have 2G or no data



Citizen / family

- Speak or type in Hindi / English
- Aadhaar + OTP verified account
- See beds, doctors, blood live
- SMS when there is no internet



Hospital

- Accept referral, reserve a bay
- AI pre-arrival (SBAR) note
- Update every 108 minutes
- Named Data Update Officer



108 / 102 control room

- Signed incident, no re-asking
- Nearest suitable unit
- Live status back to family
- 102 Janani for pregnancy

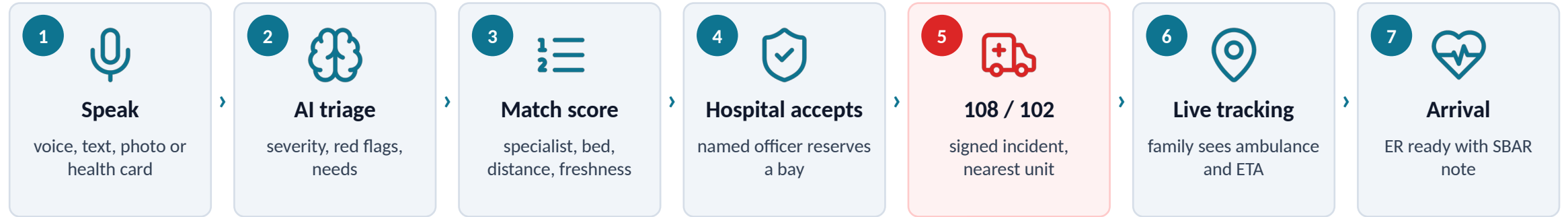


MEDREACH ENGINE

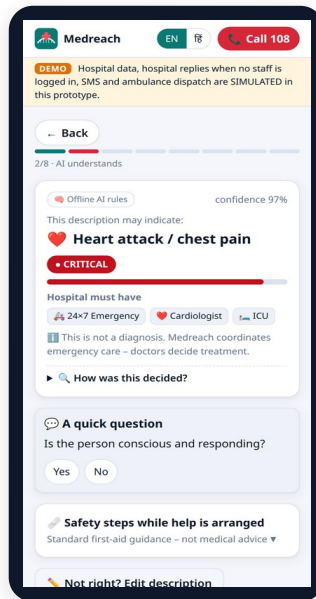
AI triage · Match score · Live capacity · 108 hand-off · SMS gateway · Tamper-evident audit

Running today: 16 hospitals across Bhopal, Raisen, Sehore and Vidisha · 9-ambulance practice 108 fleet · English and Hindi

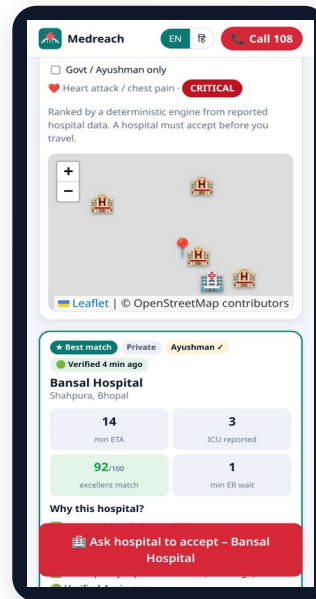
From First Words to a Confirmed Bed in 7 Steps



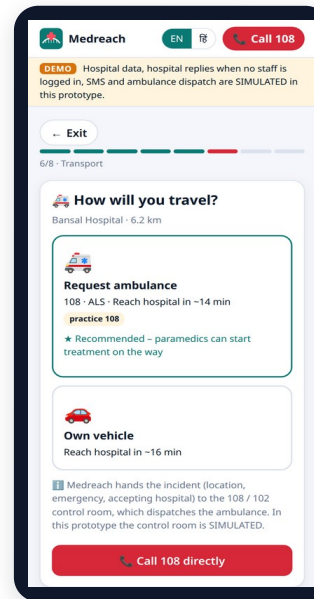
Speak or pick an emergency



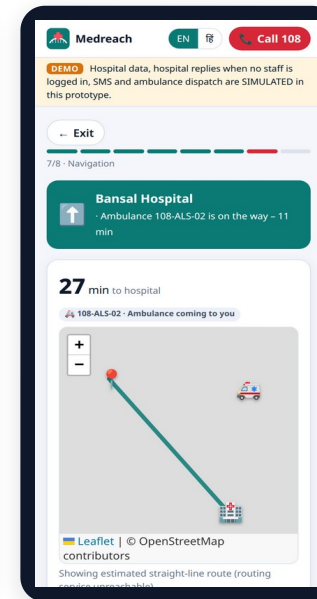
AI understands – in seconds



Match score: 92/100



Accepted → choose 108



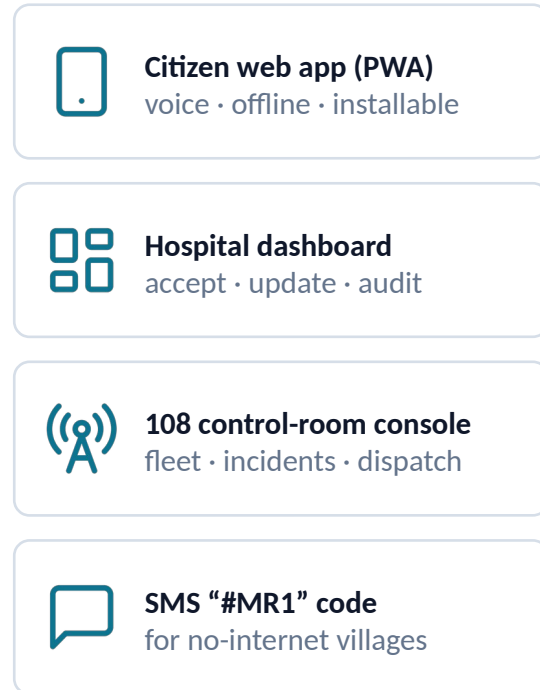
108 unit tracked live

RULES WE NEVER BREAK

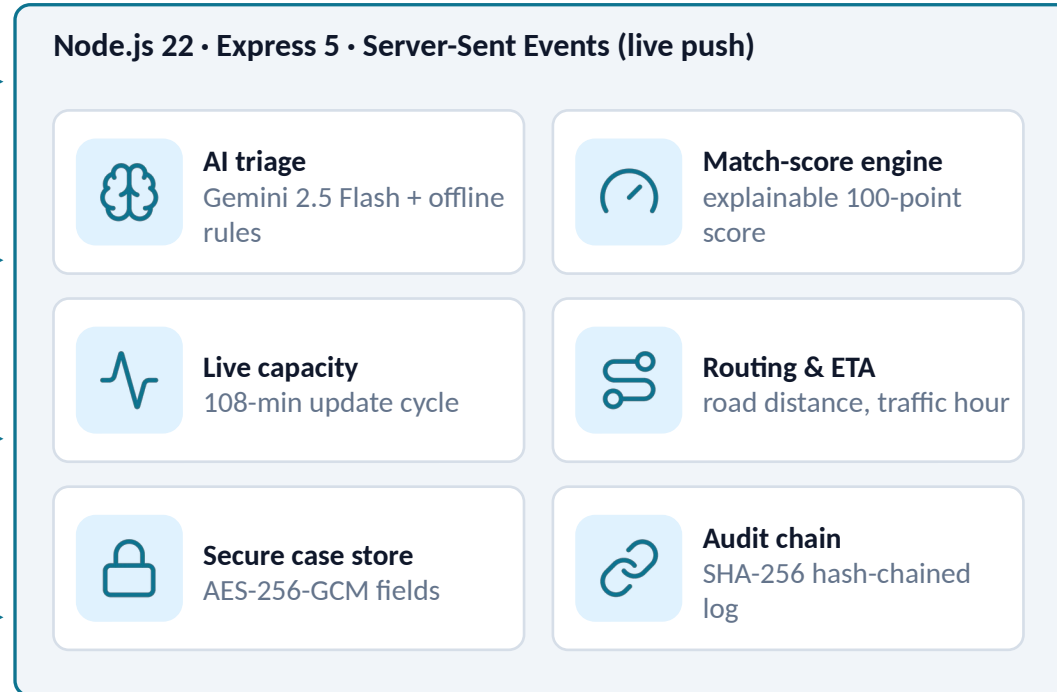
- Hospital accepts before anyone travels
- Only hospitals that can treat the case
- 60 s without answer → next hospital
- Never diagnoses: “may indicate”
- Call 108 visible on every screen

Architecture: Simple, Secure, Integrable

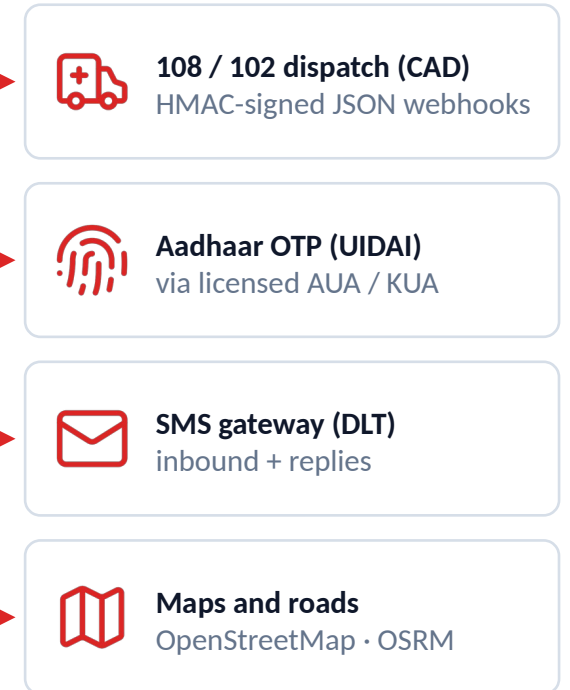
CHANNELS



MEDREACH SERVER



INTEGRATIONS

**SECURITY BY DESIGN**

Aadhaar never stored (keyed hash + last 4) · OTP limits & lockout · signed 108 messages with replay window · rate limits · strict CSP · per-field staff permissions

**OPEN, LOW-COST STACK**

Web Speech API (voice) · Tesseract.js (health-card OCR on the phone) · Leaflet maps · Service Worker (offline) · runs on one small cloud server (Render)

Built, Deployed and Tested – Not a Mockup

94

automated tests
passing

3

connected portals:
citizen, hospital, 108

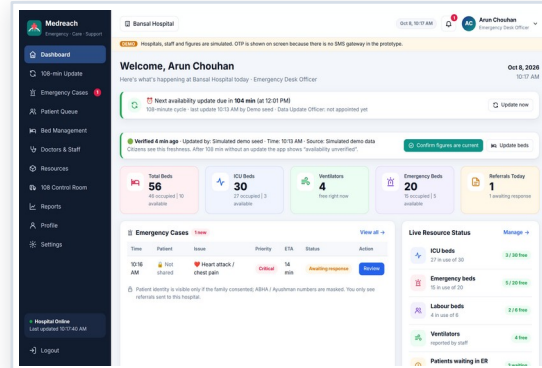
16

hospitals • 4 districts

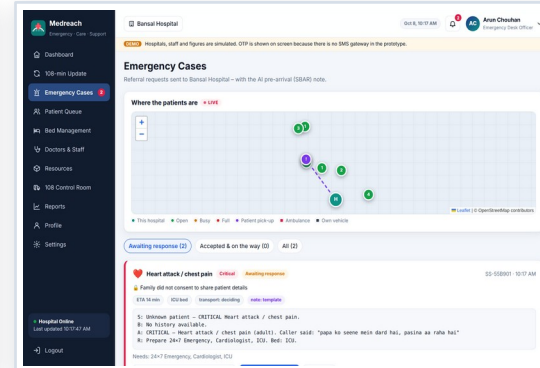
9

ambulances in practice
108 fleet

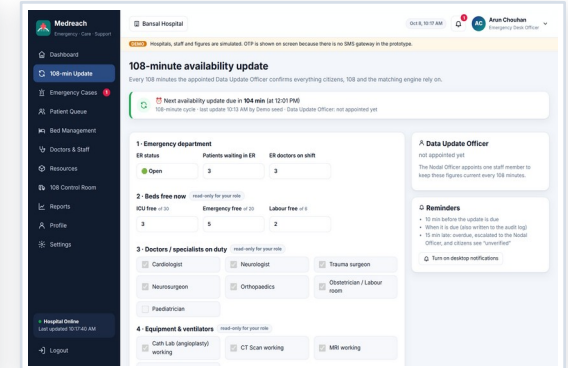
LIVE NOW
medreach-2.onrender.com



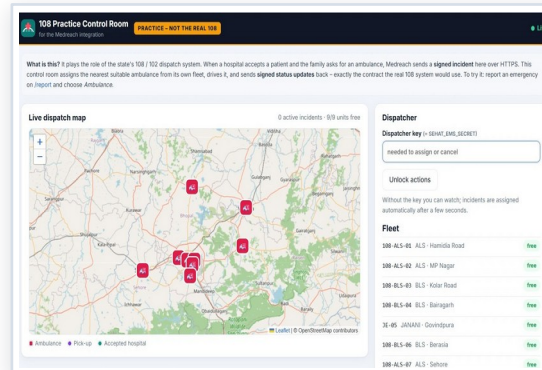
Hospital dashboard – live capacity



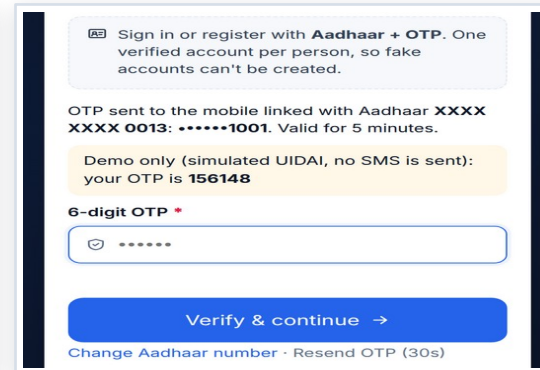
Incoming referral + AI SBAR note



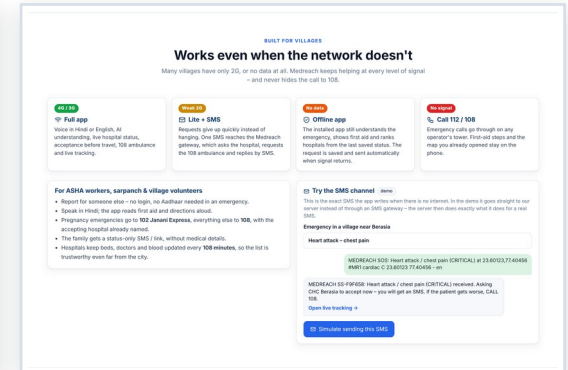
108-minute update & Data Officer



108 control room (live site)



Aadhaar + OTP sign-in



Villages: SMS when no internet

What Makes Medreach Different

Capability	Calling around	Map apps	Medreach
Knows if the specialist & ICU are free now	if someone answers	✗	✓
Hospital confirms before you travel	sometimes	✗	✓
Speak in Hindi / Hinglish, AI understands	✓	✗	✓
108 gets location + accepting hospital	✗	✗	✓
Works on 2G, SMS and offline	✓	✗	✓
Data owner + 108-minute freshness	✗	✗	✓
Aadhaar-verified users, tamper-evident audit	✗	✗	✓

Medreach does not replace 108 or hospitals – it connects them.



Explainable match score

Each hospital gets a score out of 100 with its parts shown.



Accept before travel

No family leaves without a “yes” and a bay.



108-minute update cycle

Named officer, reminders, “unverified” if late.



Signed 108 hand-off

HMAC contract any state CAD can adopt.



#MR1 SMS code

Whole request fits in one SMS on 2G.



AI pre-arrival note

SBAR summary reaches the ER first.

Impact: Minutes Saved Are Lives Saved

60 s

a hospital must answer, or the next one is asked

108 min

maximum age of hospital data before it is flagged

4

network levels covered: 4G, 2G + SMS, offline, call

0

Aadhaar numbers stored – only a keyed reference



Citizens & families

- Right hospital the first time
- No form-filling in panic: just speak
- Family tracks the ambulance live
- Villages served on 2G / SMS



Hospitals

- Patients arrive announced, bay ready
- SBAR note before arrival
- Fewer unsuitable walk-ins
- Audit log protects the staff



108 / Health department

- No re-asking: complete incident
- Live map of capacity statewide
- Data owner for every hospital
- Planning from real demand

SCALE-UP

Pilot Bhopal division – 16 hospitals, 4 districts



Madhya Pradesh all districts, with the state 108 operator



India any state's 108 / 102, same open contract

Why Medreach Deserves to Win



It works today

Live website: speak → hospital accepts
→ 108 ambulance → tracking, end to end.



Ready for real 108

Signed incident contract, proven with
our practice control room.



Built for rural MP

Hindi voice, 2G / SMS / offline, 102
Janani Express for mothers.



Secure & private

Aadhaar OTP, AES-256-GCM, tamper-
evident audit, no Aadhaar stored.



Low cost, scalable

Open-source stack, one web app for
every phone, AI optional.



Responsible AI

Never diagnoses; hospitals decide; Call
108 always one tap away.

SCAN TO TRY IT LIVE



medreach-2.onrender.com

Code: [github.com/
prakharpurwal518-collab/Medreach](https://github.com/prakharpurwal518-collab/Medreach)

Thank you
धन्यवाद